Abdelrahman Eissa

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Education

York University – Lassonde School of Engineering Toronto, ON
B.Eng., Spec. Hons. Engineering ***GPA: 9.0 / 9.0*** *Expected Graduation, 2028*

Experience

Software Engineer Intern June 2025 – Aug. 2025
BWell (Health-Tech Startup)

- Designed and implemented the core diagnostic module of an agentic workflow that automatically triages patients.
- Architecture inspired by Microsoft's *AI Clinician* paper achieving 100% accuracy on in-house and physician-curated test cases.
- Deployed pipeline using Python, FastAPI, and LLM orchestration tools

Peer Mentor June 2025 – Present
York University *Toronto, ON*

- Provided academic mentorship for MATH 1028 and MATH 1025 during the summer term.
- Currently serving as a mentor for MATH 1028, supporting students with course material and academic transitions.

Competitions

York Engineering Competition (YEC) | *Programming* 2024

- Secured 3rd place in Programming for an optimization tool designed to improve Google Maps transit routing.
- Calculated optimal arrival times to significantly minimize waiting intervals between transit transfers.

York Engineering Competition (YEC) | *Junior design* 2024

- Won 1st place in Junior Design by designing a structure capable of withstanding earthquakes while supporting a 5kg load.
- Represented the university at the Ontario Engineering Competition (OEC) as a result of the Junior Design victory.

Projects

E-Commerce Website | *Next.js, React, Stripe, NextAuth, SQLite*

- Developed a full-stack e-commerce application using Next.js and React for a responsive user experience.
- Integrated Stripe API for secure payment processing and NextAuth for Google authentication.

Multilayer Perceptron(MLP) from Scratch | *C++*

- Architected a Multilayer Perceptron (MLP) from the ground up using C++ without external ML libraries.
- Implemented fundamental neural network components including the forward pass and backpropagation algorithms.
- Validated the model against the MNIST dataset, achieving a 96% classification accuracy.

Automatic Assignment Marker | *Java, Google Vision API, Spring Boot*

- Engineered an automated marking system using Java and Spring Boot to streamline the grading process.
- Integrated the platform with Google Classroom API to maintain existing teacher workflows.
- Incorporated AI to interpret student intent during grading, increasing marking accuracy to 98%.
- Reduced student feedback latency by eliminating manual grading bottlenecks.

Custom 3D Game Engine | *C++, OpenGL*

- Developed a rendering engine using OpenGL and implemented the Bresenham line-drawing algorithm for pixel manipulation.
- Programmed 3D-to-2D projection and texture mapping to render complex spatial environments.
- Created a custom math library to handle matrix multiplications for camera rotations and coordinate transformations.

Awards

- Top in World – Edexcel O-Level Mathematics (British Council)
- Top in Country – Cambridge O-Level Combined Science (British Council)

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL, HTML/CSS
Libraries/Tools: Google Vision API, Flask, FastAPI, React, Git, Docker, LLM orchestration, Machine Learning, Game Development